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Efficient Link Capacity and QoS Design for Wormhole Network-on-Chip

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Abstract

The allocation of link capacities is an important phase in the automated design process of a network on chip based system. Communication resources must be efficiently allocated so that Quality of Service requirements are met but total network cost is minimized.

The paper contributions are twofold: First we introduce a novel analytical delay model for virtual channelled wormhole networks with non-uniform link capacities that eliminates costly simulations in the inner-loop of the optimization process. Second, we present an efficient capacity allocation algorithm that assigns link capacities so that packets of each flow arrive within the specified time. Using examples, we demonstrate the benefit of capacity allocation for a typical SoC, where the traffic is heterogeneous and delay requirements may largely vary.

1. Introduction

As the density of VLSI designs increases, the number of modules in a typical System on Chip (SoC) is expected to be significantly larger than today. Traditional interconnection schemes might no longer be adequate for these systems. The concept of *Network on a Chip* (NoC), introduced in [1], suggests that different modules would be connected by a simple network of shared links and routers, which would enable them to exchange information using data packets. As was analyzed in [2], NoC power and area costs scale much better than shared bus, segmented bus or dedicated point to point connections. Moreover, [3] has made the observation that NoC introduces a better SoC design methodology as standard network interfaces eliminate the need to redesign modules every time communication needs are globally increased (as opposed to shared busses or dedicated links). Networks that employ wormhole flow-control were introduced in [4] and are in wide use in multi-computer interconnect. In wormhole networks each packet is divided into a sequence of *flit*s which are transmitted over physical links one by one in pipeline fashion. A hop-to-hop credit mechanism assures that a flit is transmitted only when the receiving port has free space in its input buffer. Wormhole routing based networks fit the chip interconnect challenge as they minimize communication latencies, require small buffer space and are simple to implement.

In [5], Bolotin et al. presented the QNoC architecture: a wormhole based network which guarantees quality of service (QoS) of different inter-module traffic types. Another important feature is the ability to construct an asymmetric network with variable speed inter-router links.

NoC is customized through an automatic network design phase for a particular SoC implementation ([3], [5], [6]). The goal of the QNoC design process is to minimize the network cost (in terms of area and power) while maintaining the required QoS for the specific system communication demands ([5]). This customization is based on adequate bandwidth allocation to each of the network links. Insufficient allocation will not meet performance requirements while lavish allocation will result in excessive power and area consumptions.

Previous NoC literature dealt with the network performance mainly by simulations as no adequate analytical model that allows quick assessment of network performance exists. Existing models of wormhole networks ([7]-[14]) have not dealt with both heterogeneous traffic patterns and virtual channels, which are fundamental characteristics of NoC interconnect. The use of simulations makes the task of searching for a low cost and high performance capacity allocation computationally extensive and it does not scale well with the size of the problem. On the other hand, a detailed exact analytical solution of a complex NoC is intractable. Using exclusively simplistic approximations may lead to a fast but inaccurate design.

In this work, we propose a hybrid methodology to the network design problem that combines the best of both worlds. First, we propose a novel and simple analytical delay model for a wormhole based NoC, which approximates the network behavior in a wide range of loads. Given any system (in terms of topology, routing and link capacities) and its communication demands (in terms of packets length and generation rate), the model estimates the delay experienced by every source destination pair. To the best of our knowledge, this is the first modeling and analysis of a wormhole network with non-uniform link capacities. Next, we propose an algorithm that applies the delay modeling and analysis to efficiently allocate capacities to network links without need for repetitive simulations in the inner-loop of optimization ([5]). Simulation runs are only used for final verification and fine tuning of the system. Using a design example we demonstrate that our algorithm considerably decreases the total NoC cost and significantly improves the quality and speed of the customization process.

The rest of this paper is organized as follows: Section 2 reviews related work, Section 3 briefly discusses the QNoC architecture, Section 4 describes our wormhole delay model and derives the appropriate equations, Section 5 presents our capacity allocation algorithm, Section 6 provides a design example and simulation results. Finally, section 7 concludes the paper.